

ANCHORAGE MERRILL FIELD, AK, USA

WMO: 702735

Lat: 61.217N

Lon: 149.855W

Elev: 138

StdP: 14.62

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26409

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-6.0	-16.1	2.3	-7.6	-11.6	3.0	-3.2	18.0	29.8	15.4	30.1	1.7	80	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	74.0	59.8	71.5	58.7	68.5	57.2	61.5	71.4	60.0	68.5	58.7	66.3	6.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.2	70.2	63.0	55.7	66.4	62.5	54.7	63.9	61.8	27.3	71.5	26.4	69.0	25.5	65.9	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.8	12.4	10.6	DB	-13.8	79.4	5.1	3.6	-17.5	81.9	-20.5	84.0	-23.3	86.0	-27.0	88.6	
			WB	-14.2	64.1	5.0	3.1	-17.9	66.3	-20.8	68.1	-23.6	69.9	-27.3	72.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	16.2	21.0	26.0	38.9	49.7	57.6	61.0	58.5	50.1	36.8	23.0	18.1	
	DBStd	18.13	12.72	10.51	9.21	6.26	5.05	4.34	3.81	3.86	5.15	8.12	10.32	10.78	
	HDD50	5275	1048	811	743	335	66	1	0	0	60	411	811	989	
	HDD65	9809	1513	1231	1208	784	473	225	135	204	446	875	1261	1454	
	CDD50	960	0	0	0	1	57	230	341	265	65	1	0	0	
	CDD65	19	0	0	0	0	0	4	11	4	0	0	0	0	
	CDH74	96	0	0	0	0	1	29	49	17	0	0	0	0	
	CDH80	7	0	0	0	0	0	3	4	0	0	0	0	0	
Wind	WSAvg	3.9	3.0	3.2	4.0	4.2	5.0	5.0	4.5	3.9	3.6	3.5	3.3	3.1	
Precipitation	PrecAvg	16.4	0.8	0.8	0.7	0.5	0.7	1.1	1.8	2.8	3.0	1.9	1.2	1.1	
	PrecMax	27.6	2.7	2.4	2.8	1.9	1.9	3.4	4.5	9.8	7.7	4.3	2.8	2.7	
	PrecMin	8.1	0.0	0.1	0.0	0.0	0.0	0.2	0.2	0.3	0.7	0.4	0.0	0.1	
	PrecStd	3.5	0.6	0.5	0.5	0.5	0.4	0.7	1.0	1.8	1.7	0.9	0.8	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	44.5	47.5	59.1	72.4	77.9	79.2	76.3	66.1	55.9	47.5	46.4	
		MCWB	37.5	37.2	38.9	46.4	53.8	60.7	62.3	61.5	54.1	47.3	39.5	38.4	
	2%	DB	39.4	39.4	43.5	54.7	66.8	73.0	74.5	72.2	63.0	51.9	41.3	39.2	
		MCWB	35.2	35.1	36.5	43.4	51.4	58.2	61.1	59.9	53.2	45.0	37.5	35.2	
	5%	DB	35.8	36.5	40.6	52.1	63.0	69.8	71.6	69.1	60.6	49.4	38.4	35.1	
		MCWB	33.3	33.9	34.8	41.8	49.4	56.8	59.6	58.3	52.0	43.9	35.6	32.9	
	10%	DB	32.4	34.3	37.8	49.3	59.5	66.2	68.8	66.2	57.8	47.1	36.1	32.3	
		MCWB	30.5	32.1	33.2	40.4	47.8	55.0	58.2	57.0	51.0	43.1	33.9	30.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	37.8	40.2	48.1	55.2	62.4	64.0	63.1	56.1	49.5	40.6	38.8
			MCDB	43.7	41.7	45.8	57.3	69.9	75.4	76.6	73.7	62.0	53.0	45.3	44.7
2%		WB	35.5	36.1	37.4	44.7	52.3	59.5	61.9	61.1	54.5	46.6	38.2	35.5	
		MCDB	38.9	38.9	42.0	52.5	63.8	70.6	72.4	69.5	59.9	49.5	41.3	38.7	
5%		WB	33.1	34.2	35.5	42.6	50.6	57.6	60.4	59.6	53.4	45.1	35.9	32.8	
		MCDB	35.7	36.6	39.6	50.4	60.9	67.0	69.3	66.7	58.1	48.3	38.2	34.7	
10%		WB	30.5	32.1	33.8	41.0	48.9	56.1	59.1	58.3	52.3	43.5	33.8	30.7	
		MCDB	32.3	34.3	37.4	48.4	58.1	65.0	66.9	64.6	56.7	47.0	35.8	32.2	
Mean Daily Temperature Range	5% DB	MDBR	11.7	13.2	15.8	15.5	15.5	14.2	12.6	12.9	12.5	11.1	10.7	11.1	
		MCDBR	12.6	12.6	15.1	19.4	21.3	19.7	18.1	18.0	16.0	12.5	10.6	13.4	
		MCWBR	9.2	9.2	9.8	11.0	10.8	9.7	8.7	8.8	8.8	7.5	7.8	10.0	
	5% WB	MCDBR	12.2	12.0	13.6	17.5	18.4	17.4	16.1	15.1	13.2	10.2	10.0	12.2	
		MCWBR	9.3	9.0	9.3	10.6	9.9	9.0	8.2	8.2	7.7	6.6	7.5	9.6	
Clear-Sky Solar Irradiance	taub	0.233	0.286	0.305	0.346	0.367	0.358	0.371	0.356	0.317	0.296	0.241	N/A		
	taud	2.128	2.233	2.267	2.248	2.314	2.421	2.415	2.463	2.558	2.471	2.126	N/A		
	Ebn at Noon	160	220	258	267	271	276	269	264	258	221	150	N/A		
	Edh at Noon	12	21	28	35	36	33	32	29	22	16	11	N/A		
All-Sky Solar Radiation	RadAvg	91	309	822	1360	1638	1688	1500	1187	773	359	132	47		
	RadStd	13	43	88	120	147	108	130	129	107	43	20	6		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	+2.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	+3.60	+3.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page